CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	Subclause(s)	Tracking No.	Year
IEC 60601-1: 2005 (ed 3 .0)	8.9	2017	2015
Category			
MED			
Subject	**Keywords**	**Developed by**	**Approved at**
Creepage distance	Creepage distance Working voltage, PCB traces	ETF-3	2016 CTL Plenary Meeting

Question

How should the creepage distance be measured in a scenario where the point with the highest
working voltage is behind a point with a lower working voltage?

See the case study below for details:

Creepage distance requirement on a Class II switch mode power supply unit designed for 2 MOPP.

A working voltage of 240 Vrms was measured between the secondary side and the nearest printed track on the primary side.

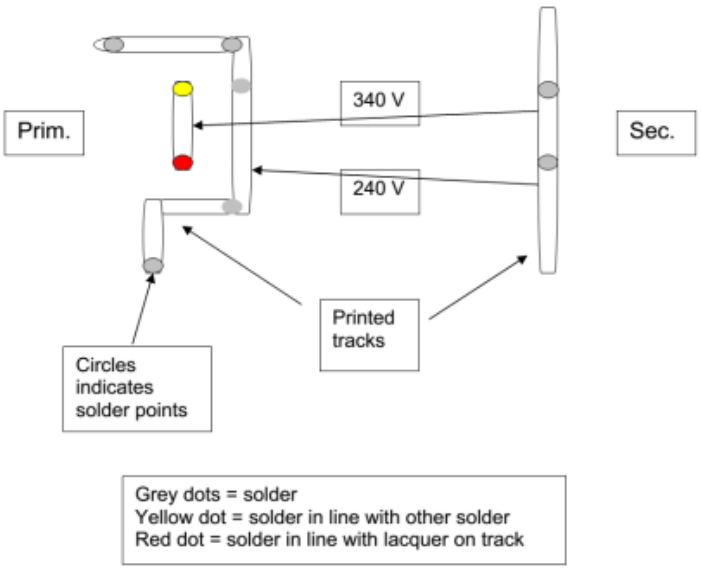
However, behind the nearest printed track on the primary side there is a point with a working voltage of 340 Vrms.

The creepage distance between secondary and the nearest 240 V point is 8 mm, which meets the requirement for 240 V.

The creepage distance between the 340 V point and the 240 V point is 1 mm.

Question:

How shall the creepage distance between primary and secondary be measured in such a case?



Decision

Both scenarios 1) and 2) below must be considered.

1) Where the tracking path travels over the lacquer/solder mask and directly to the secondary trace, the creepage distance must be measured directly from the 340 V trace to the secondary trace. In this case > 9 mm would be reported, i.e. 1 mm + the width of the trace + 8 mm > 9 mm.
(Using interpolation, Table 12 leads to a required 10.7 mm).

2) The effect of short circuiting between 340 V and 240 V must be considered.
a) If after short circuiting, the voltage of 240 V trace raises to 340 V, as if the 240 V trace were a dead metal part, the creepage distance is counted as 1 mm + 8 mm = 9 mm (as per clause 5.2.2.8 of IEC 60664-1:2007). In this case the 9 mm would be reported versus the required 10.7 mm.
b) If, however, the effect of short circuiting between the 340 V trace and the 240 V trace results in no increase in voltage on the 240 V trace, the creepage distance from the 240 V trace to the secondary is based on 240 V working voltage, leading to a required 8 mm according to Table 12. In this case the value in scenario 1) would also be reported for the 340 V separation.

Explanatory notes

There are two general scenarios to be considered:

- one scenario where the lacquer/solder mask over the 240V trace acts as an insulation barrier,
  and
- the other scenario where the lacquer/solder mask easily breaks down.

Note that the lacquer/solder mask cannot be relied upon as isolation, however it cannot be assumed that it will not act as insulation, hence both scenarios must be considered